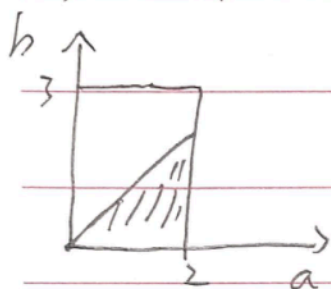


$$(1) (A) \Omega = \{(0,0), (0,1), \dots\} \quad \#\Omega = 4 \times 3 = 12$$

$$\text{其中 } A = \{(1,0), (2,0), (3,0), (2,1), \text{划掉}(2,2), (3,1), (3,2)\} \quad \#A = 6$$

$$P(A) = \frac{\#A}{\#\Omega} = \frac{1}{2}$$

$$(13) \text{ 设实根} \Leftrightarrow (2a)^2 - 4b^2 < 0 \Leftrightarrow a^2 < b^2 \quad \Omega = [0, 2] \times [0, 3]$$



$$P(B) = \frac{\text{划掉} \int_B}{\int_{\Omega}} = \frac{1}{3}$$

$$\text{划掉} A = \text{划掉} \Omega$$

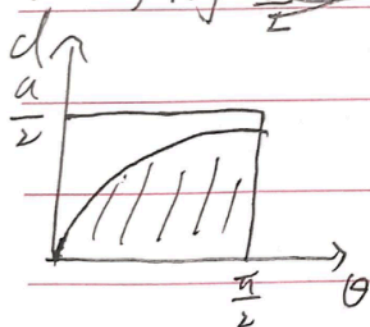
(2) 针的位置和角度应为均匀分布

设针的中点离平行线的最近距离为 $d \in [0, \frac{a}{2}]$

与平行线所成夹角为 $\theta \in [0, \frac{\pi}{2}]$

$$\Omega = \{(d, \theta) \mid d \in [0, \frac{a}{2}], \theta \in [0, \frac{\pi}{2}]\}$$

相交, 则 $\frac{1}{2} \sin \theta \leq d < \frac{1}{2} \sin \theta$ 划掉



$$P(A) = \frac{\int_0^{\frac{\pi}{2}} \frac{1}{2} \sin \theta d\theta}{\frac{a}{2} \cdot \frac{\pi}{2}} = \frac{2}{\pi}$$

$$(3) P(A) = \frac{n!}{N^n}$$

$$P(B) = \frac{C_n^n \cdot n!}{N^n}$$

(4) $A = \{\text{取到的数为奇数}\}$ $3p + p = 1 \quad p = \frac{1}{4}$
 $B = \{\text{取到的数不超过50}\}$ $C = \{\text{取到的数超过50}\}$
 $P(A) = P(A|B)P(B) + P(A|C)P(C) = \frac{13}{200}$

(5) $\Rightarrow$: $\forall x \in \Omega$, 如果 $x \in A$, 则 $x \in B$, 则 $I_A(x) = I_B(x) = 1$
 若 $x \notin A$, 则 $I_A(x) = 0 \leq I_B(x)$

$\Leftarrow$: $\forall x \in \Omega$, $I_A(x) \leq I_B(x)$ 有三种可能:

$I_A(x) = I_B(x) = 1$	$x \in A, x \in B$
$I_A(x) = 0, I_B(x) = 1$	$x \notin A, x \in B$
$I_A(x) = I_B(x) = 0$	$x \notin A, x \notin B$

$\therefore$ 若 $x \in A$, 则 $x \in B \quad \therefore A \subset B$

(13) 只需由数学归纳法, 只需证两个集合的情形,

若 $x \in A_1 \cup A_2$, 则 $I_{A_1 \cup A_2}(x) = 1$ 而 $I_{A_1}(x), I_{A_2}(x)$ 中至少有一个是1

若 $x \notin A_1 \cup A_2$, 则 $I_{A_1 \cup A_2}(x) = I_{A_1}(x) = I_{A_2}(x) = 0$

$\therefore I_{A_1 \cup A_2}(x) = \max I_{A_i}(x)$

若 $x \in A_1 \cap A_2$, 则 $I_{A_1 \cap A_2}(x) = 1$, 且 $I_{A_1}(x) = I_{A_2}(x) = 1$

若 $x \notin A_1 \cap A_2$, 则 $I_{A_1 \cap A_2}(x) = 0$, $I_{A_1}(x), I_{A_2}(x)$ 至少有一个为0

$\therefore I_{A_1 \cap A_2}(x) = \min I_{A_i}(x)$